

# Installing SkyWarden on Windows

## 1. Install Python

1. Download Python 3.10 or newer from <https://www.python.org/downloads/windows/>
2. In the installer, tick **"Add python.exe to PATH"**, then click Install.
3. Check it worked. Open **Command Prompt** and run:

```
python --version
```

(If `python` isn't found, try `py --version` and use `py` instead of `python` below.)

## 2. Put the files in a folder

Create a folder, for example `C:\SkyWarden`, and copy these two files into it:

- `skywarden.py`
- `requirements.txt`

### 3. Install the libraries

In Command Prompt:

```
cd C:\SkyWarden
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
```

This installs PyQt6 (the window), requests, ephem (target altitude) and pywin32 (classic ASCOM).

### 4. Run it

```
cd C:\SkyWarden
venv\Scripts\activate
python skywarden.py
```

Always start it from this folder. It saves

`config_skywarden.json` and `skywarden_log.txt` in the folder you run it from.

On launch it scans for devices automatically. You can rescan any time from the **Hardware & Configuration** tab.

### 5. Install your hardware drivers (as needed)

- **ASCOM:** install the ASCOM Platform from <https://ascom-standards.org>, then your equipment's ASCOM driver. Installed drivers show up in the dropdowns after a scan.
- **Alpaca:** the Alpaca server (on this PC or another machine on your network) must be running.
- **INDI:** an INDI server must be running on port 7624.

## 6. First-time setup

1. Open **Hardware & Configuration**.
2. Click **Scan**, then choose your mount, dew heater and cloud sensor.
3. Set the **Dew Heater Switch ID** to match your heater's switch number.
4. Check your latitude/longitude on the dashboard.
5. Click **Save Settings**.

**Test with simulators before real use.** Install the ASCOM Telescope and Switch simulators, press **Force Run Verification**, then try the emergency button while watching the log. Do this before you trust it with real equipment.

## Troubleshooting

| Problem                                | Fix                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <code>python</code> is not recognised  | Reinstall Python and tick "Add to PATH", or use <code>py</code> instead of <code>python</code> .                                           |
| No module named PyQt6 (or similar)     | Activate the venv ( <code>venv\Scripts\activate</code> ), then run <code>pip install -r requirements.txt</code> again.                     |
| No devices found                       | Allow Python through Windows Firewall on your <b>private</b> network (needed for Alpaca discovery). INDI is only found on your own subnet. |
| "needs Windows and pywin32" in the log | Run <code>pip install pywin32</code> inside the venv.                                                                                      |
| ASCOM "Class not registered" error     | The driver may be 32-bit only. Install 32-bit Python and repeat steps 3 and 4.                                                             |

| Problem                                       | Fix                                                                               |
|-----------------------------------------------|-----------------------------------------------------------------------------------|
| Safety checks fail with "weather unavailable" | The PC needs internet access, or untick "Enable Internet API Weather Validation". |